

6. (c) $-y \log(\alpha^T x) - (1-y) \log(1 - \alpha^T x)$ for $y \in \{0,1\}$

$$\begin{aligned}\nabla f(x) &= -y \nabla (\log(\alpha^T x)) - (1-y) \nabla \log(1 - \alpha^T x) \\ &= -y \frac{1}{\alpha^T x} \alpha + (1-y) \frac{1}{(1 - \alpha^T x)} \alpha\end{aligned}$$

$$x_{k+1} = x_k + \epsilon_k \nabla f(x_k)$$

(i) choose an initial $x(0)$,

(ii) start the k from 0, and for each iteration increase by ϵ .

(iii) compute the gradient for each iteration

(iv) ~~now~~ check if the current value is less or greater than the previous value. And find the optimum value of k .

6. (e) $\|y - Ax\|_2^2 + \|x\|_2^2$

$$\begin{aligned}\nabla f(x) &= \nabla \left((y - Ax)^T (y - Ax) \right) + \nabla \|x\|_2^2 \\ &= -2A^T (y - Ax) + 2x\end{aligned}$$

$$x_{k+1} = x_k + \epsilon_k \nabla f(x_k)$$

Same gradient descent/ascent method above applies

$$\frac{\partial \log(L(\theta))}{\partial \theta} = \eta/2 - 1/2 \theta^T X_i X_i^T$$

$$\frac{\partial}{\partial \theta} = 0$$

$$n = \theta^T X_i X_i^T$$

$$I = \theta \frac{X_i X_i^T}{n}$$

$$\theta^{-1} = \frac{X_i X_i^T}{n} \Rightarrow \underline{\underline{\Sigma = \frac{X_i X_i^T}{n}}}$$

5. convex, strictly convex, concave or none:

a. $\max(0, 1 - y * \alpha^T x)$ for $y \in \{\pm 1\}$ $\Rightarrow$ not convex not concave

This looks like maximum-margin classification, which is convex

b. $\log(1 + e^{\alpha^T x}) \Rightarrow$ convex

Second derivative is positive

c. $-y \log(\alpha^T x) - (1-y) \log(1 - \alpha^T x)$ for $y \in \{0, 1\}$

$\Rightarrow$ convex

(1) ~~The~~ $\log(\alpha^T x)$: concave, $-y \log(\alpha^T x) \Rightarrow$ convex

(2) $\log(1 - \alpha^T x)$: concave, $-(1-y) \log(1 - \alpha^T x) \Rightarrow$ convex

Sum of (1) & (2) is convex.

d. $\|y - Ax\|_1 \Rightarrow$ convex

$\|u\|_1 = |u_1| + |u_2| + \dots + |u_n| \rightarrow$ L1 norm (convex)

$\|y - Ax\|_1$ is the L1 norm of $y - Ax$, it is convex

e. $\|y - Ax\|_2^2 + \|x\|_2^2 \Rightarrow$ convex

L2 norm
convex

strictly
convex
(L2 norm)

3.b. Let $x, y \in \mathbb{R}^n$ Find eigenvalues & eigenvector of xy^T

M is $n \times n$ matrix, $M \in \mathbb{R}^{n \times n}$, $M = xy^T$

$Mv = \lambda v$, where v is eigenvector and λ is eigenvalues of M

$$(xy^T)v = \lambda v$$

$$x(y^T v) = \underbrace{v}_{\text{eigenvector}} \underbrace{\lambda}_{\text{eigenvalue}}$$

x is the eigenvector, while $y^T x$ is the eigenvalue.

3.c. Let A be a real symmetrical $p \times p$ matrix with eigenvalues $\lambda_1, \dots, \lambda_p$. Show that $\sum_{i=1}^p \sum_{j=1}^p a_{ij}^2 = \sum_{i=1}^p \lambda_i^2$.

$$A = V \Lambda V^{-1}$$

$$A^2 = V \Lambda V^{-1} V \Lambda V^{-1}$$

$$A^2 = V \Lambda^2 V^{-1} \Rightarrow \boxed{V^{-1} A^2 V = \Lambda^2} \quad \text{get trace of matrix}$$

$$(i) \operatorname{tr}(V^{-1} A^2 V) = \operatorname{tr}(A^2) = \sum_{i=1}^p a_{ii}^2 \quad (\text{similarity trace})$$

$$(ii) \operatorname{tr}(\Lambda^2) = \lambda_1^2 + \lambda_2^2 + \dots + \lambda_p^2 \Rightarrow \sum_{i=1}^p \sum_{j=1}^p a_{ij}^2 = \sum_{i=1}^p \lambda_i^2$$

4.a. Suppose $\{x_i\}_{i=1}^n$ are n IID samples from a p -variate normal distribution with known mean 0 and unknown covariance Σ (positive definite). Show that $\hat{\Sigma} = \frac{X^T X}{n}$ is the MLE of Σ , where $X = \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{pmatrix}$

(i) Joint likelihood:

$$N(X | \mu, \Sigma) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} e^{-\frac{1}{2} (X - \mu)^T \Sigma^{-1} (X - \mu)}$$

$$L(\Sigma | X) = \prod_{i=1}^n \Rightarrow (2\pi)^{-np/2} |\Sigma|^{-n/2} e^{-\frac{1}{2} \sum_{i=1}^n x_i^T \Sigma^{-1} x_i}$$

(ii) log likelihood:

$$\log(L(\Sigma)) = -np/2 \log(2\pi) + n/2 \log |\Sigma| - \frac{1}{2} \sum x_i^T \Sigma^{-1} x_i$$

$$\Sigma^{-1} = \Theta$$

$$\text{cov}(\tilde{X}) = I$$

$$\text{cov}(\tilde{X}) = E\left[\left((Ax+b) - E[Ax+b]\right)\left((Ax+b) - E[Ax+b]\right)^T\right]$$

$$E[Ax+b] = AE[x] + b = A\mu + b \quad E[x] = \mu$$

$$\text{cov}(\tilde{X}) = E\left[(Ax+b - A\mu - b)(Ax+b - A\mu - b)^T\right]$$

$$= E\left[(Ax - A\mu)(Ax - A\mu)^T\right]$$

$$= A E\left[(x - \mu)(x - \mu)^T\right] A^T$$

Σ given for matrix X

$$\text{cov}(\tilde{X}) = A \Sigma A^T = I \quad (\text{Asked to be } I = \text{cov}(\tilde{X}))$$

$$\Rightarrow A = \Sigma^{-1/2}$$

mean = 0

$$E[x] = \mu \quad E[\tilde{X}] = 0 \quad \tilde{X} = Ax + b$$

$$E[\tilde{X}] = A E[x] + b$$

$$E[\tilde{X}] = A\mu + b \Rightarrow b = -A\mu$$

↪

$$\tilde{X} = Ax + b = Ax - A\mu$$

$$= A(x - \mu)$$

$$\tilde{X} = \Sigma^{-1/2} (x - \mu)$$

(c) Show that the OLS prediction $\hat{y} = X\hat{\beta}_{OLS}$ can be written as $\hat{y} = Hy$ for some matrix H .

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

$$\hat{y} = X \hat{\beta}_{OLS}$$

$$= \underbrace{X (X^T X)^{-1} X^T}_{\text{matrix } H} y$$

$$\boxed{\hat{y} = Hy}$$

3 (a) $X \in \mathbb{R}^r$, known mean $E(X) = \mu$, known covariance $\text{cov}(X) = \Sigma$, where Σ is positive definite.

Find an affine transformation of X such that the transformed variable, $\tilde{X}$, is such that $E(\tilde{X}) = 0$ and $\text{cov}(\tilde{X}) = I$

transform
 X : mean; $E(X) = \mu$ covariance; $\text{cov}(X) = \Sigma$ (positive def.)
 $\rightarrow \tilde{X}$: mean; $E(\tilde{X}) = 0$ covariance; $\text{cov}(\tilde{X}) = I$ (identity)

$$\text{cov}(X) = E[(X - \mu)(X - \mu)^T] \quad E[X] = \mu$$

$$\text{cov}(\tilde{X}) = E[(\tilde{X} - E[\tilde{X}])(\tilde{X} - E[\tilde{X}])^T]$$

$$\tilde{X} = AX + b$$

(b) Calculate $E(\hat{\beta}_{OLS})$ and $cov(\hat{\beta}_{OLS})$

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y, \quad y = X\beta + \epsilon$$

$$\begin{aligned} (*) E(\hat{\beta}_{OLS}) &= E[(X^T X)^{-1} X^T y], \quad \left\{ E(y) = X\beta \right\} \\ &= (X^T X)^{-1} X^T E(y) \\ &= (X^T X)^{-1} X^T E[X\beta + \epsilon] \\ &= (X^T X)^{-1} X^T E(X\beta) + E(\epsilon) \quad \text{given } E(\epsilon) = 0, \text{ given} \\ &= \underbrace{(X^T X)^{-1} X^T X}_{I} \beta \end{aligned}$$

$$E(\hat{\beta}_{OLS}) = \beta$$

$$\begin{aligned} (*) cov(\hat{\beta}_{OLS}) &= cov[(X^T X)^{-1} X^T y] \\ &= (X^T X)^{-1} X^T \underbrace{cov(y)}_{\sigma^2 I} \underbrace{(X^T X)^{-1} X^T}_{X(X^T X)^{-1}}^T, \quad cov(y) = \sigma^2 I \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \end{aligned}$$

$$cov(\hat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1}$$

Homework 0

(2) $y = X\beta + \varepsilon$ $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $p < n$, $\beta \in \mathbb{R}^p$, $\varepsilon \in \mathbb{R}^n$

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} \|y - X\beta\|_2^2,$$

Assume ε , $E(\varepsilon) = 0$, $\operatorname{cov}(\varepsilon) = \sigma^2 I$

(a) $\hat{\beta}_{OLS}$?

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} \|y - X\beta\|_2^2$$

$$\frac{\partial \|y - X\beta\|_2^2}{\partial \beta} = 0$$

$$\begin{aligned} \|y - X\hat{\beta}\|_2^2 &= (y - X\hat{\beta})^T (y - X\hat{\beta}) \\ &= y^T y - \underbrace{y^T X \hat{\beta}}_{[1 \times n] [n \times p] [p \times 1]} - \underbrace{\hat{\beta}^T X^T y}_{[1 \times p] [p \times n] [n \times 1]} + \hat{\beta}^T X^T X \hat{\beta} \\ &\quad \xrightarrow{\frac{\partial \hat{\beta}^T X^T y}{\partial \hat{\beta}} = X^T y} \end{aligned}$$

$$= y^T y - 2\hat{\beta}^T X^T y + \hat{\beta}^T X^T X \hat{\beta}$$

$$\frac{\partial \|y - X\hat{\beta}\|_2^2}{\partial \hat{\beta}} = 0 = -2X^T y + 2X^T X \hat{\beta}$$

$$(X^T X)^{-1} / 2X^T X \hat{\beta} = 2X^T y / (X^T X)^{-1}$$

$$(X^T X)^{-1} (X^T X) \hat{\beta} = (X^T X)^{-1} X^T y$$

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$